

Monitoring Consciousness across Anaesthesia and Disorders of Consciousness using Metrics of Critical Brain Dynamics

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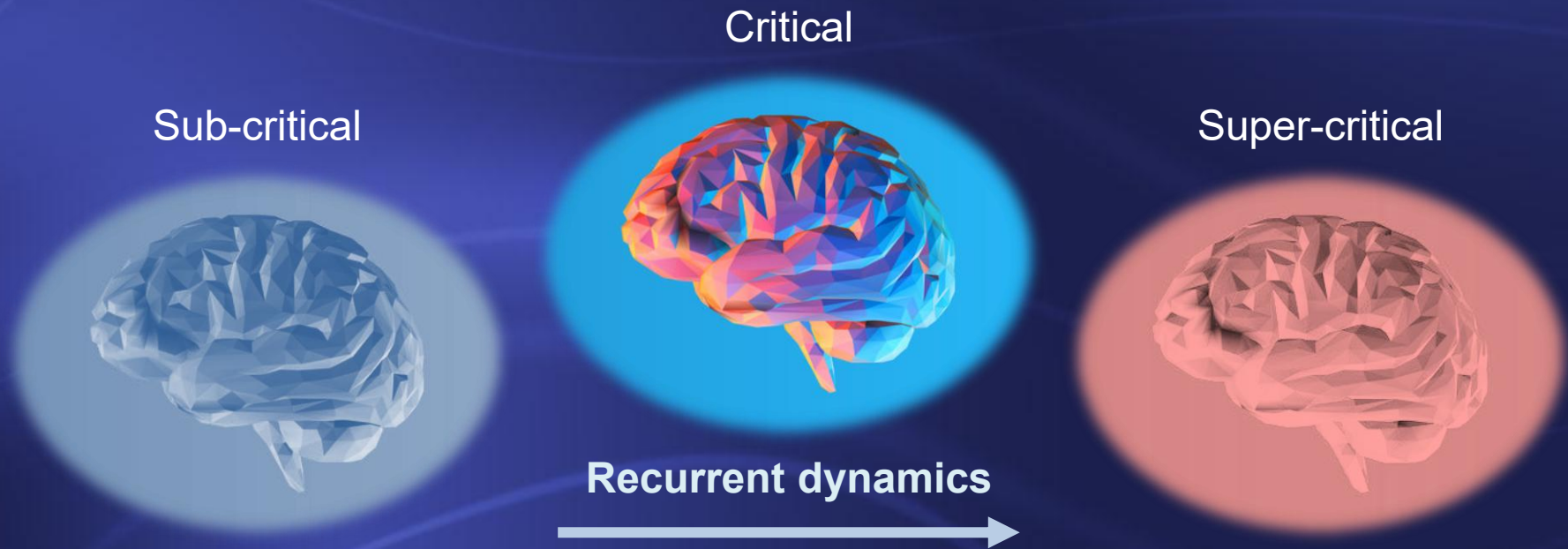
Ben-Gurion University of the Negev, Israel

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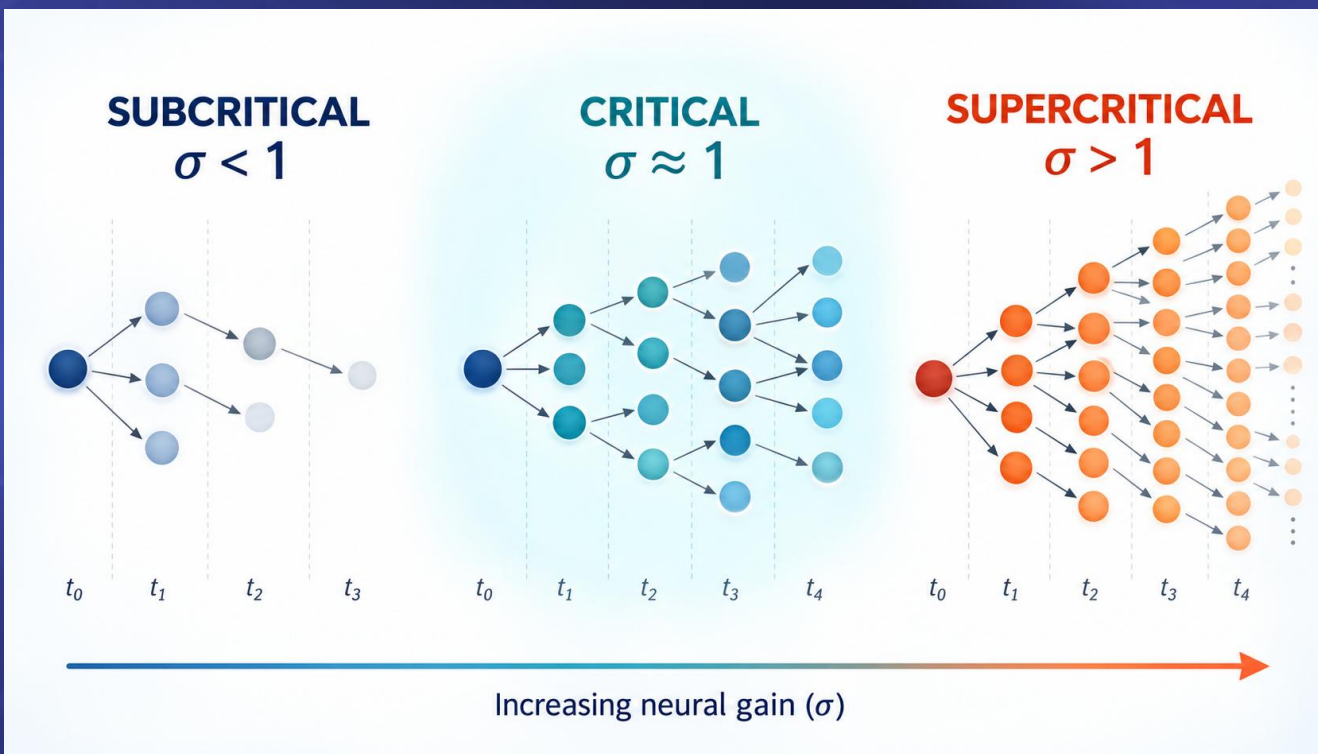
Critical Brain Dynamics: A Computational Operating Regime

- The healthy brain is hypothesized to operate near critical transitions between dynamical regimes
- Computational advantages: Sensitivity and communication, Richness and flexibility (entropy), Integration across space and time...
- Departures from this regime may therefore provide markers of altered consciousness and brain dysfunction.



The Critical Brain and Avalanches

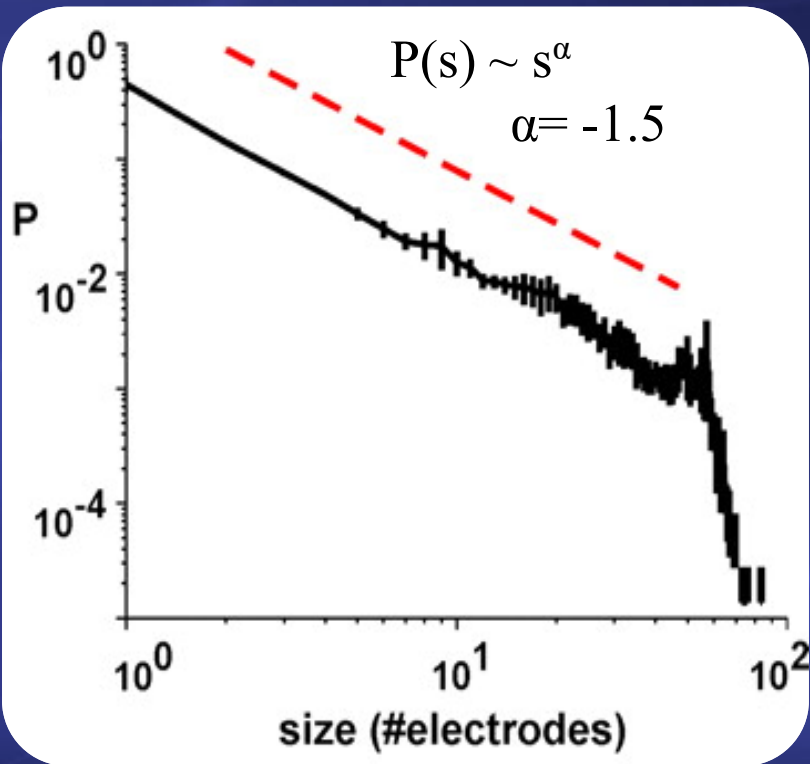
- A major line of research focuses on the transition between premature termination of activity propagation and exponential blow-up.
- The theory of **branching processes** predicts that in critical systems ($\sigma=1$) the size distribution of activity cascades, **neuronal avalanches**, would be a **power law**, $P(S) \propto S^{-3/2}$.



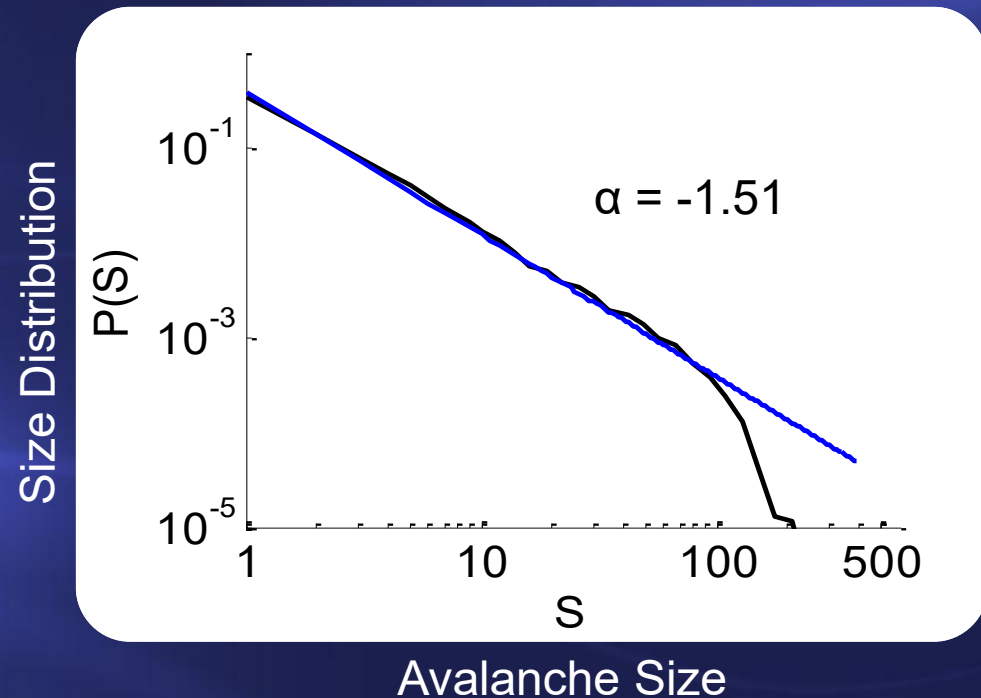
Empirical Evidence for Criticality in the Brain

Neuronal Avalanches

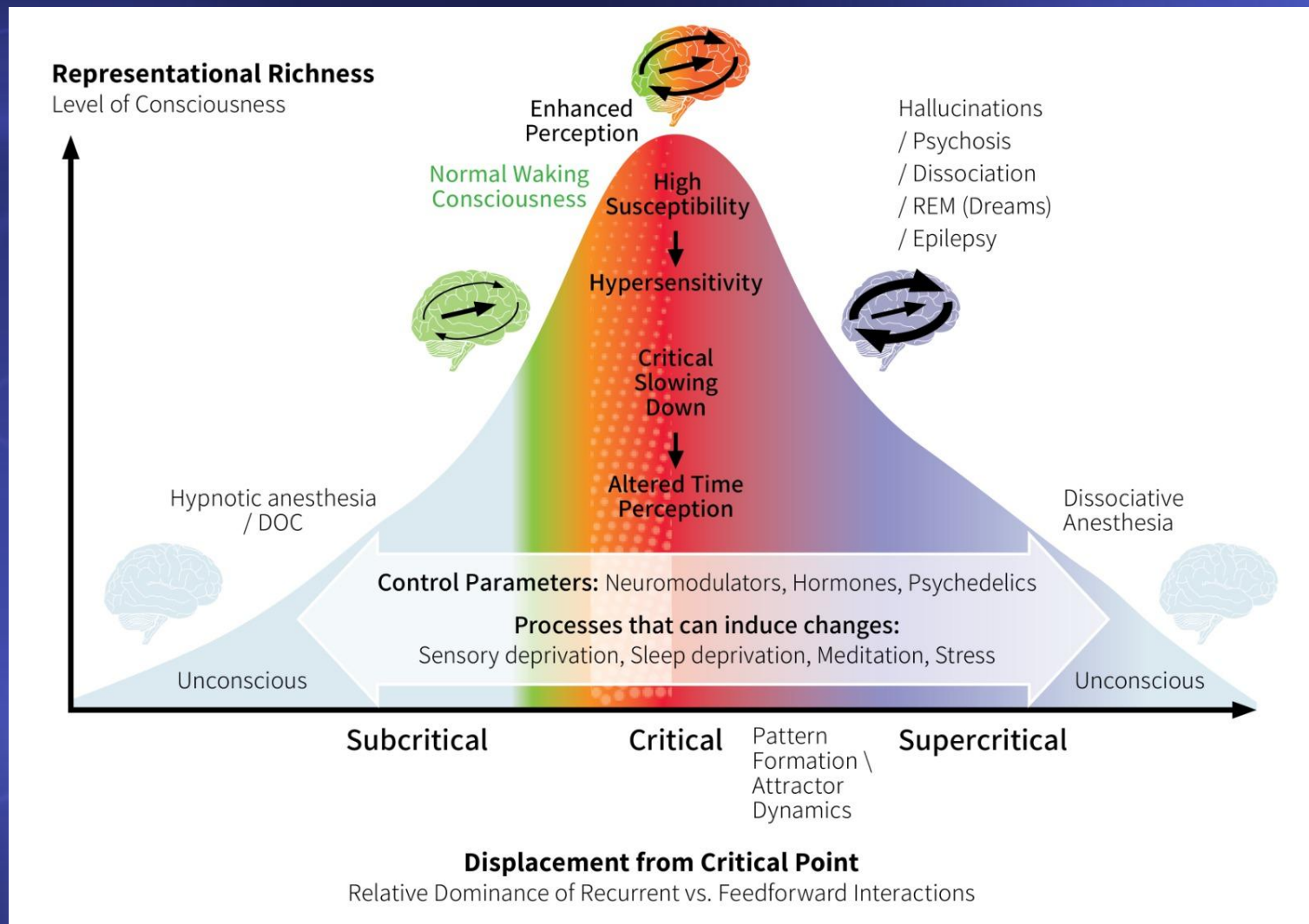
Organotypic cultures



Human MEG



The *ConCrit* Framework: Consciousness and Criticality



The *ConCrit* Framework: A Theoretical Perspective

Leading theories of consciousness (ToCs) require neural dynamics that support:

- Integration and differentiation
- Global propagation and long-timescale integration
- Susceptibility and flexible state transitions
- Metastability and rich representational repertoire

Near-critical dynamics naturally combine these properties

→ Therefore, criticality aligns with diverse theories of consciousness and may provide a common dynamical substrate for conscious processing.

The *ConCrit* Framework: An Empirical Perspective

- Conscious wakefulness is generally associated with **near-critical** dynamics.
- Loss of consciousness often involves **displacement from criticality**; recovery involves a return toward it.
- **Hypnotic anesthesia** (Propofol, Isoflurane...), deep NREM sleep, and DOC commonly show **subcritical** features.
- **Dissociative anesthetics**, particularly ketamine, may produce region- and dose-dependent shifts, including shifts toward **supercritical** regimes.
- Seizures can reflect **supercritical** propagation.
- **Psychedelics** often increase dynamical diversity and **proximity to criticality**.

The direction of change depends on the critical transition, metric, brain region, and perturbation.

Spontaneous EEG Criticality Tracks Consciousness, Not Simply Responsiveness

communications biology

Article



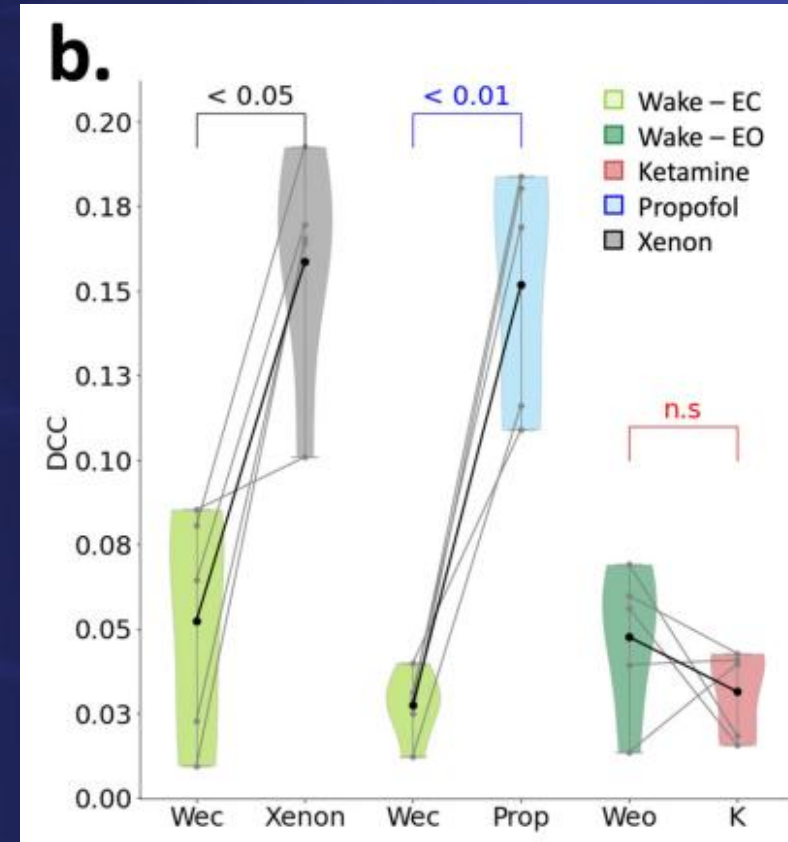
<https://doi.org/10.1038/s42003-024-06613-8>

Critical dynamics in spontaneous EEG predict anesthetic-induced loss of consciousness and perturbational complexity

Check for updates

Charlotte Maschke^{1,2,3,13}, Jordan O'Byrne^{3,4,13}, Michele Angelo Colombo⁵, Melanie Boly⁶, Olivia Gosseries^{7,8}, Steven Laureys^{7,9,10}, Mario Rosanova⁵, Karim Jerbi^{3,4,11} & Stefanie Blain-Moraes^{1,12}

- Participants were behaviorally unresponsive under propofol, xenon, and ketamine.
- Propofol and xenon unconsciousness were associated with displacement from avalanche criticality and the edge of chaos.
- Under ketamine, vivid experience was retained and critical dynamics were relatively preserved.
- Resting-state criticality metrics predicted independently measured PCI.



higher DCC = farther from criticality

Propofol Anesthesia Shifts Avalanche Dynamics Toward a Subcritical Regime

Baseline

mean = 0.40

median = 0.40

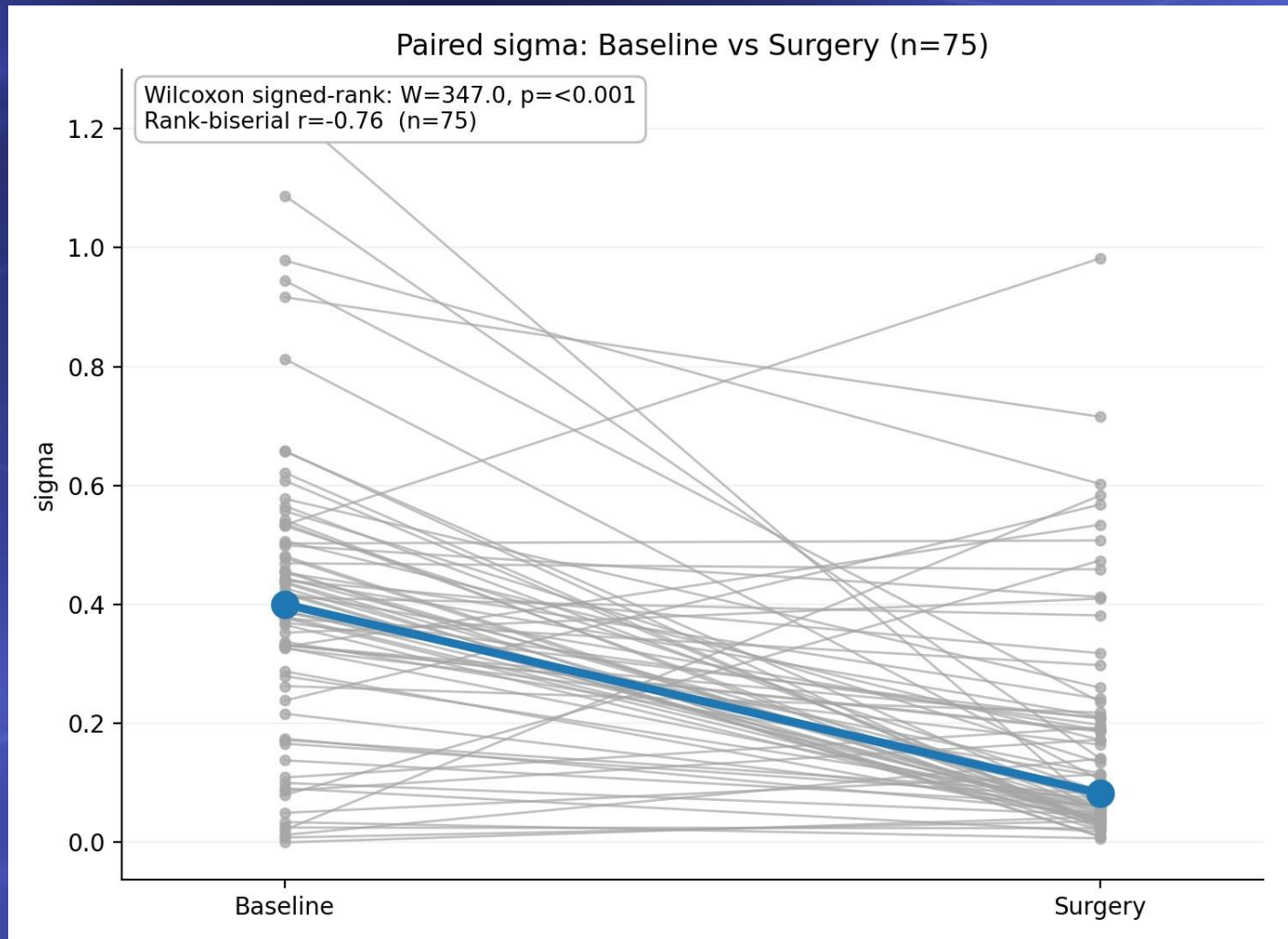
SD = 0.25

Surgery

mean = 0.17

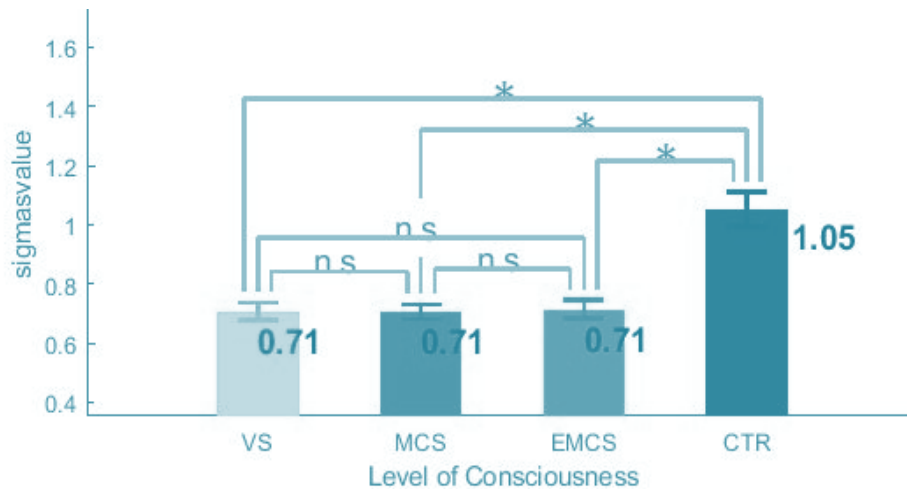
median = 0.08

SD = 0.19



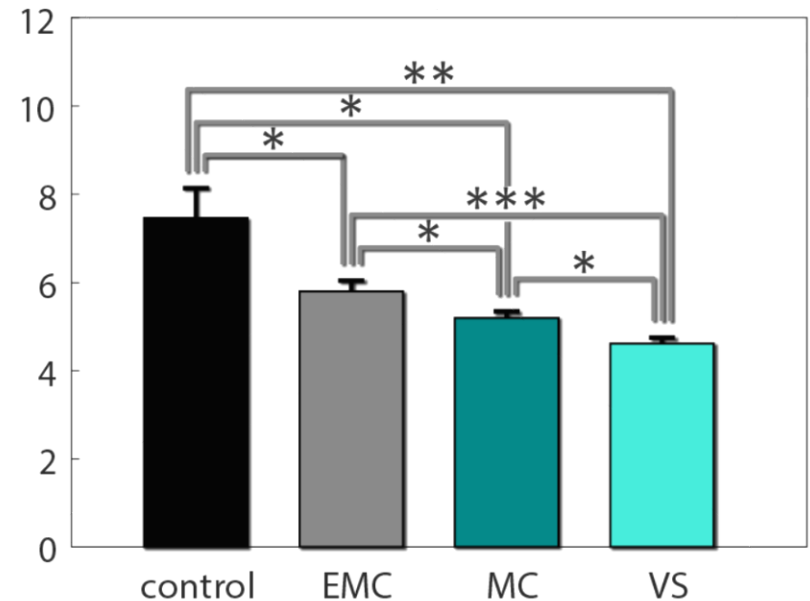
Criticality Metrics Capture Complementary Aspects of DOC

Gain Metric σ



Disorders of Consciousness

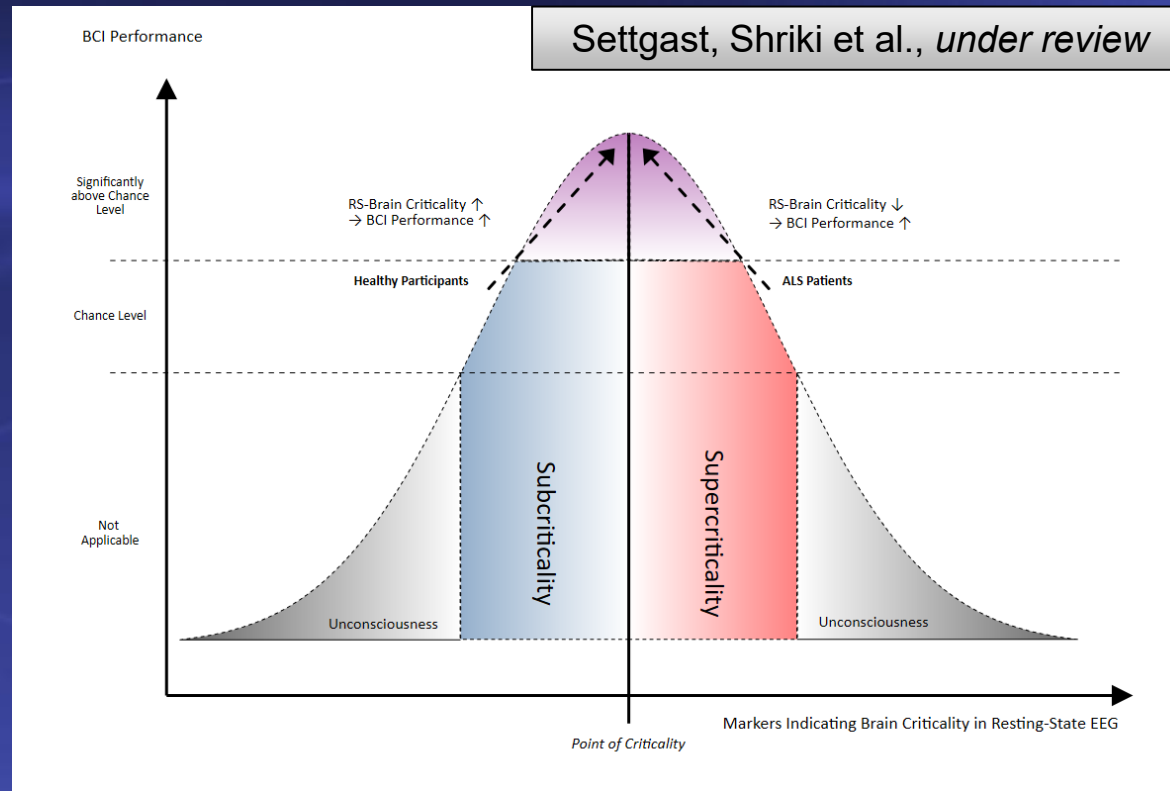
Multi-scale criticality metric



Disorders of Consciousness

EEG Data from the lab of Jacobo Diego Sitt,
INSERM, Paris, France

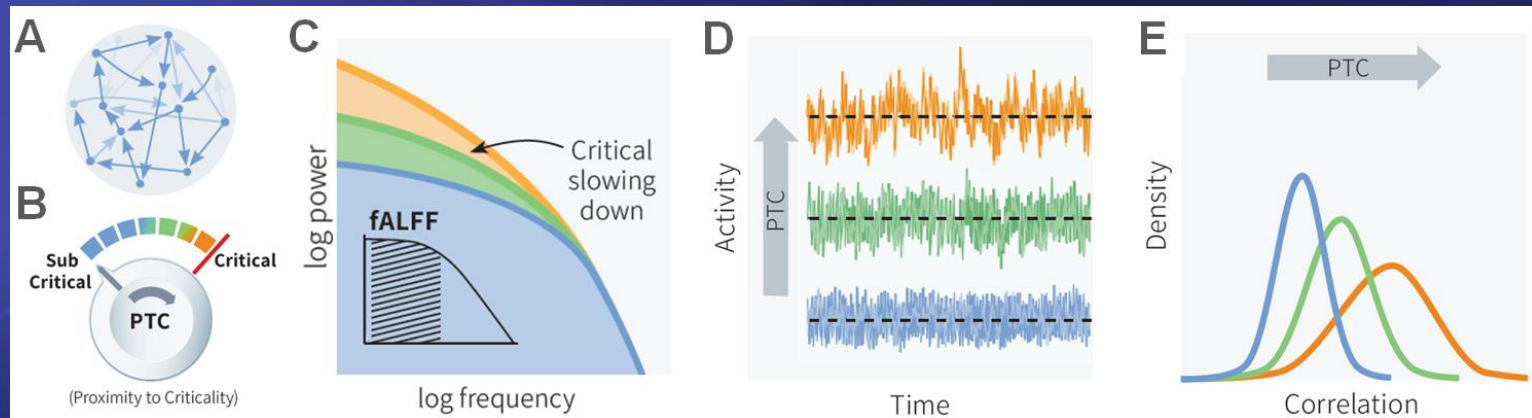
Criticality May Index Readiness for BCI Communication



- Proximity to criticality fluctuates over time.
- BCI performance improves as dynamics approach the critical regime.
- Controls generally approached criticality from the subcritical side, whereas ALS patients often approached it from the supercritical side.
- Pharmacological modulation may shift brain dynamics toward criticality, potentially improving BCI performance

Criticality Across Cortical Hierarchies

- **Our finding:** Estimated proximity to criticality **increases along cortical hierarchies**, with the **DMN closest to criticality**.
- **Interpretation:** Near-critical dynamics may support long-timescale integration underlying self-related and narrative thought.
- **Anesthesia context:** Higher-order network integration, including the DMN, is disrupted even when lower-level sensory connectivity remains relatively preserved.
- **Prediction:** Loss and recovery of consciousness may involve changes in the cortical gradient of criticality—not simply a uniform brain-wide shift. **DMN**



fMRI Data: HCP, 7T, 40 subjects, 5-min rest, 360 ROIs (180 per hemisphere)

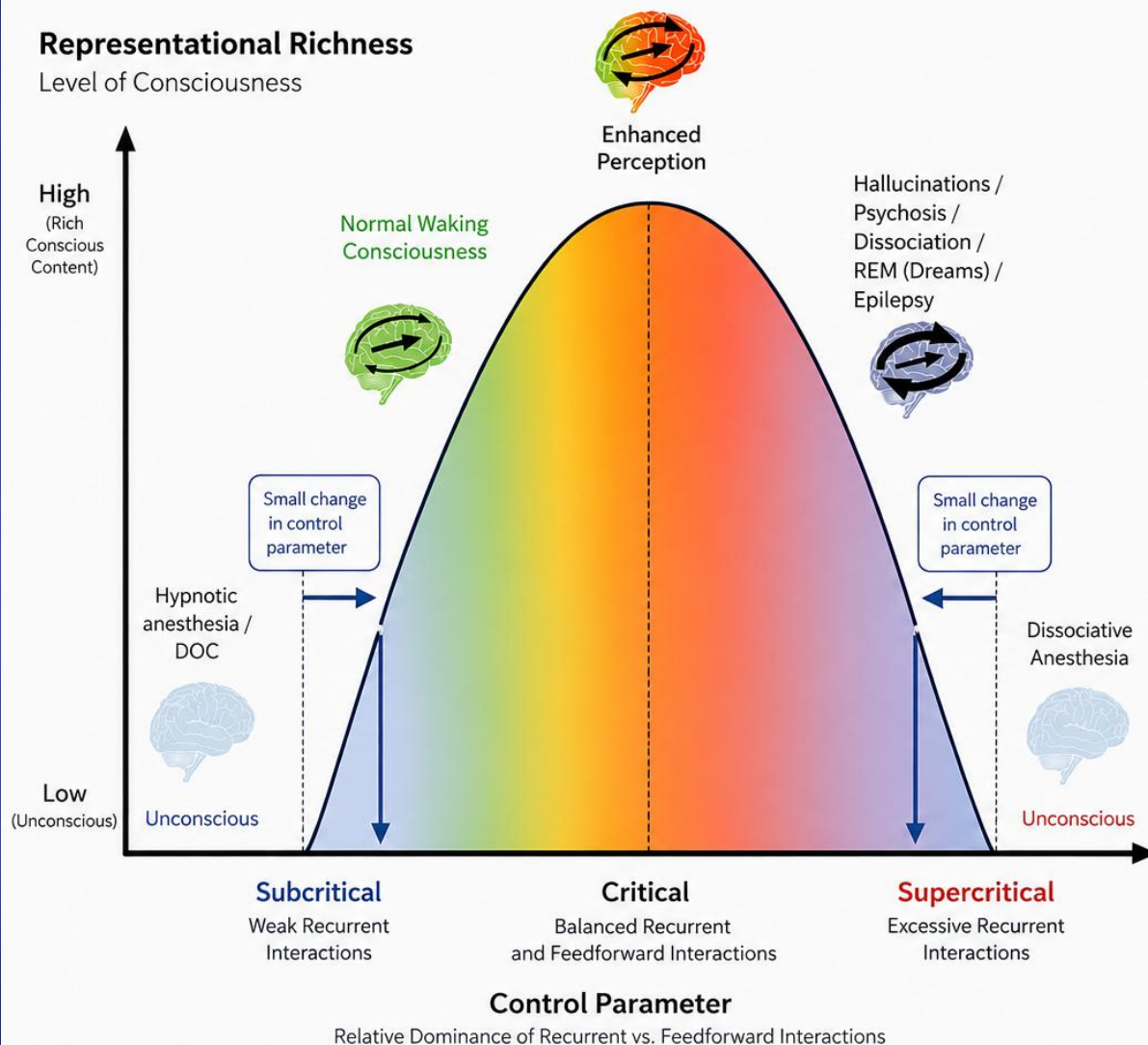
Criticality-Informed BCI

Passive and Active Measures Answer Complementary Questions

Approach	Question answered	Main limitation
Spontaneous criticality	Is the brain in a regime capable of flexible, integrated responses?	Cannot demonstrate specific conscious content or intention
Evoked-response BCI	Does the brain discriminate or process the stimulus?	Response may occur without conscious access
Active/task BCI	Can the patient follow a command through volitional modulation?	Nonresponse may reflect an unfavorable state or task-related impairment
Combined approach	When is a task result most informative?	Requires prospective, individualized validation

Use criticality as a passive state marker to optimize testing windows, adapt BCI parameters, and interpret negative results.

Continuous vs. Binary Transitions



Continuous changes, but abrupt collapses

- Brain dynamics vary gradually across many states (e.g., sleep, psychedelics).

Sharp drops can occur far from the critical point

- Small changes in the control parameter— even far from criticality— can cause a sharp loss of representational richness and consciousness.

Core idea

- The brain operates on a nonlinear landscape where robustness can give way to abrupt transitions under small perturbations.

From Neuromarker to Clinical Monitor

- **Validate the target:** Do criticality metrics track consciousness rather than drug effects or behavioral responsiveness?
- **Demonstrate added value:** Does criticality improve assessment beyond standard EEG markers and BCI performance?
- **Track individual changes:** Can monitoring capture within-patient fluctuations and recovery?
- **Establish bedside reliability:** Are estimates robust with few electrodes, short recordings, and clinical artifacts?
- **Guide BCI assessment:** Can criticality identify favorable testing windows and help interpret negative results?

Take-Home Messages

- Criticality metrics provide behavior-independent information about the brain's capacity for complex, integrated processing.
- They may help assess graded levels of consciousness, distinguish unconsciousness from behavioral unresponsiveness, and identify favorable windows for BCI communication.
- Clinical use will require repeated, individualized, agent- and etiology-sensitive validation—and should complement, not replace, active and perturbational BCIs.

Thanks!



Computational Psychiatry and
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